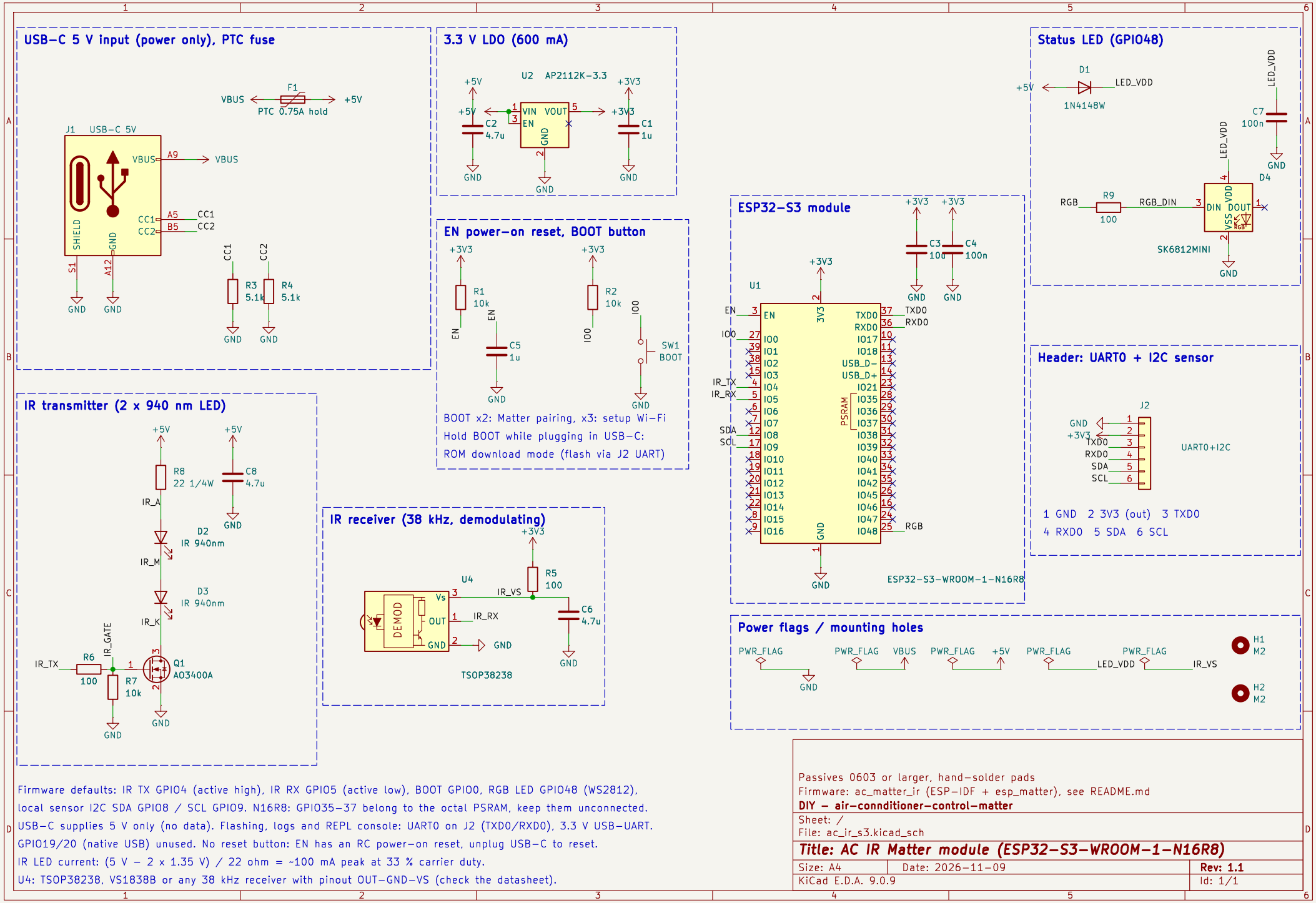

USB-C 5 V input (power only), PTC fuse
F1
VBUS
PTC 0.75A hold
+5V
J1 USB-C 5V
VBUS
A9
CC1
A5
CC2
B5
SHIELD
GND
A12
S1
R3 5.1k
R4 5.1k
3.3 V LDO (600 mA)
U2 AP2112K-3.3
VIN
VOUT
EN
C2 4.7u
C1 1u
+3V3
EN power-on reset, BOOT button
R1 10k
R2 10k
C5 1u
IO0
SW1 BOOT
BOOT x2: Matter pairing, x3: setup Wi-Fi
Hold BOOT while plugging in USB-C:
ROM download mode (flash via J2 UART)
Status LED (GPIO48)
D1
1N4148W
LED_VDD
C7 100n
D4
R9 100
RGB
RGB_DIN
DIN
DOUT
VDD
VSS
SK6812MINI
ESP32-S3 module
U1
C3 100n
C4 100n
3V3
TXD0
RXD0
IO0
IO17
IO18
USB_D-
USB_D+
IO21
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
IR_TX
IR_RX
SDA
SCL
PSRAM
ESP32-S3-WROOM-1-N16R8
Header: UART0 + I2C sensor
J2
UART0+I2C
1 GND 2 3V3 (out) 3 TXD0
4 RXD0 5 SDA 6 SCL
IR transmitter (2 x 940 nm LED)
R8 22 1/4W
C8 4.7u
IR_A
D2 IR 940nm
IR_M
D3 IR 940nm
IR_K
IR_GATE
R6 100
R7 10k
Q1 AO3400A
IR receiver (38 kHz, demodulating)
U4
R5 100
IR_VS
C6 4.7u
DEMOD
Vs
OUT
TSOP38238
Power flags / mounting holes
PWR_FLAG
VBUS
LED_VDD
IR_VS
H1 M2
H2 M2
Firmware defaults: IR TX GPIO4 (active high), IR RX GPIO5 (active low), BOOT GPIO0, RGB LED GPIO48 (WS2812),
local sensor I2C SDA GPIO8 / SCL GPIO9. N16R8: GPIO35–37 belong to the octal PSRAM, keep them unconnected.
USB-C supplies 5 V only (no data). Flashing, logs and REPL console: UART0 on J2 (TXD0/RXD0), 3.3 V USB-UART.
GPIO19/20 (native USB) unused. No reset button: EN has an RC power-on reset, unplug USB-C to reset.
IR LED current: (5 V − 2 x 1.35 V) / 22 ohm = ~100 mA peak at 33 % carrier duty.
U4: TSOP38238, VS1838B or any 38 kHz receiver with pinout OUT-GND-VS (check the datasheet).
Passives 0603 or larger, hand-solder pads
Firmware: ac_matter_ir (ESP-IDF + esp_matter), see README.md
DIY – air-connditioner-control-matter
Sheet: /
File: ac_ir_s3.kicad_sch
Title: AC IR Matter module (ESP32-S3-WROOM-1-N16R8)
Size: A4
Date: 2026-11-09
Rev: 1.1
KiCad E.D.A. 9.0.9
Id: 1/1